Secrecy and authentication —— MARS_security.pv:

Query not attacker(messageV[]) is true.

Query not attacker(messageR[]) is true.

(* **Secrecy of session-key-protected communications: application messages remain secret in both vehicle-to-RSU and RSU-to-vehicle directions.***)

Non-interference messageV is true.

Non-interference messageR is true.

(* **Strong message secrecy:** the attacker cannot distinguish executions carrying different application messages.*)

Query not attacker(IDi[]) is true.

(***Vehicle identity privacy:** the attacker cannot recover the vehicle's real identity from observable protocol messages.*)

Query inj-event(RSUAccept(g_3,rho_2,X_2,Y_2,k)) ==> inj-event(VehicleConfirm(g_3,X_2,Y_2,k)) is true.

Query inj-event(VehicleAccept(g_3,X_2,Y_2,k)) ==> inj-event(RSUResponse(g_3,rho_2,X_2,Y_2,k)) is true.

(* **Injective mutual authentication and key agreement:** each acceptance corresponds to a unique matching peer session agreeing on the service group, ephemeral public values, and session key.*)

Identity indistinguishability — MARS_identity.pv: Observational equivalence is true.

(* **Identity indistinguishability:** an active network adversary cannot distinguish which of two publicly known candidate identities participates in the modeled challenge session. *)

Two-session unlinkability — MARS_unlinkability.pv: Observational equivalence is true.

(* **Two-session unlinkability:** a passive external observer cannot distinguish two successfully completed sequential sessions of the same vehicle from one session each of two different vehicles, with credential refresh and the same public service group. *)